

Homework 5: Forward and Inverse Modeling

This homework covers material from **Module 5: Forward and Inverse Modeling** and aligns with learning outcomes **LO5, LO2, LO3**.

Related Resources: - Lectures: M5 lecture notes (Forward Modeling, Inverse Methods, Depth Estimation) - Lab: Lab 5 (Talwani Forward Modeling and Inversion) — *Essential for Problems 5.1, 5.2* - Textbook: Blakely Ch. 9, 10 - Software: NumPy/SciPy for matrix operations; pyproj for coordinate transforms

Problems

1. **Problem 5.1** (Blakely Ch. 9, Problem 1) — *Analytical/Challenging*

- Implement the Talwani method for a 2D polygonal body. Derive the formula for the gravitational attraction of a vertical line segment.
- **Cross-reference:** Lab 5, Part 1 (implementation and testing); M2/lecture notes (gravity field)
- **Difficulty:** Challenging (integration geometry; algorithmic thinking)
- **Tip:** Carefully set up coordinate system; integrate line-by-line segment; verify against simple test cases (sphere, infinite slab)

2. **Problem 5.2** (Blakely Ch. 10, Problem 2) — *Linear Algebra/Moderate*

- Set up a linear inverse problem for determining magnetization direction from magnetic anomalies. Formulate the matrix equation $\mathbf{d} = \mathbf{G}\mathbf{m}$.
- **Cross-reference:** Lab 5, Part 2; M0/lecture3_notes.qmd (matrix algebra); M3/lecture notes (magnetization)
- **Difficulty:** Moderate (linear system formulation; sensitivity analysis)
- **Tip:** Identify data vector (obs. anomalies), model vector (magnetization direction), and sensitivity/Jacobian matrix; discuss rank and conditioning

3. **Problem 5.3** (Blakely Ch. 10, Problem 4) — *Application/Challenging*

- Apply Euler deconvolution to locate the source of a magnetic anomaly. Derive the Euler equation for different structural indices ($SI = 0, 1, 2, 3$).
- **Cross-reference:** Lab 5, Part 3; M5 lecture notes, Section “Depth-to-Source Methods”; M0/lecture3_notes.qmd (partial derivatives)
- **Difficulty:** Challenging (spatial derivatives; windowing strategy; multiple solutions)
- **Tip:** Compute derivatives of total field; setup overdetermined system for each window; cluster solutions to find source position

4. **Problem 5.4** (Blakely Ch. 10, Problem 6) — *Conceptual/Moderate*

- Discuss the nonuniqueness of potential field inverse problems. Provide specific examples and explain how independent information helps resolve ambiguity.

- **Cross-reference:** M1/lecture2_notes.qmd (uniqueness theorem); Lab 5 (modeling trade-offs); Final Project (real-world application)
- **Difficulty:** Moderate (conceptual depth; synthesis with prior knowledge)
- **Tip:** Show concrete examples (same anomaly from different sources); discuss role of depth weighting, regularization, and geologic constraints

5. Problem 5.5 — *Application/Moderate*

- Use Werner deconvolution to estimate depth to a magnetic contact. Show the mathematical derivation and state key assumptions.
- **Cross-reference:** Lab 5, supplementary material; M5 lecture notes, Section “Depth Estimation”; Blakely Ch. 10
- **Difficulty:** Moderate (formulation of contact model; least-squares fitting)
- **Tip:** Assume vertical contact model; derive anomaly formula; fit to observed data using least squares; discuss limitations (assumes 2D contact)

Submission Guidelines

- Submit solutions as a PDF, Quarto document, or Jupyter notebook
- Include complete derivations, mathematical formulations, and step-by-step solutions
- Include sketches/diagrams showing model geometry, sensitivity, or solution workflow
- For computational problems, include code snippets or links to reproducible scripts
- Discuss physical interpretability of results
- Expected complexity: 6–10 pages + code/figures